Category	Component	Exact part / specification
Controller	Microcontroller	ESP32-DevKitC V4 with ESP32-WROOM-32D
	USB cable	Micro-USB data cable for programming and ESP32 power
Sensors	Temperature and humidity	DHT11 sensor module
	Light	Photoresistor + LM393
	Reservoir water level	K-0135 Sensor
Irrigation	Soil sensor and pump	M5Stack Watering Unit — capacitive soil-moisture sensor + 5V pump
	Water tubing	Tubing supplied with the M5Stack unit
	Water reservoir	200ml cup
Display	Local display	0.96″ SSD1306 OLED, 128 × 64, I²C
Power and wiring	Pump power supply	Regulated 5V 2.5A DC supply
	Power connector	Matching female DC barrel-jack to screw-terminal adapter
	Breadboard and wires	Solderless breadboard + jumper wires
Plant setup	Plant	One small live plant in a pot